

Choose your own maths adventure

Nadiah Kristensen, Nicholas Buttle



The Room (Fireproof Games)

1. What did we do?



(1) — (2) — (3) — (4) — (5) — (6) — (end)

Part 4

Consider

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

Question: Find the projection of $\mathbf{b}$ onto the columnspace of A by solving $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ and $\mathbf{p} = A \hat{\mathbf{x}}$.

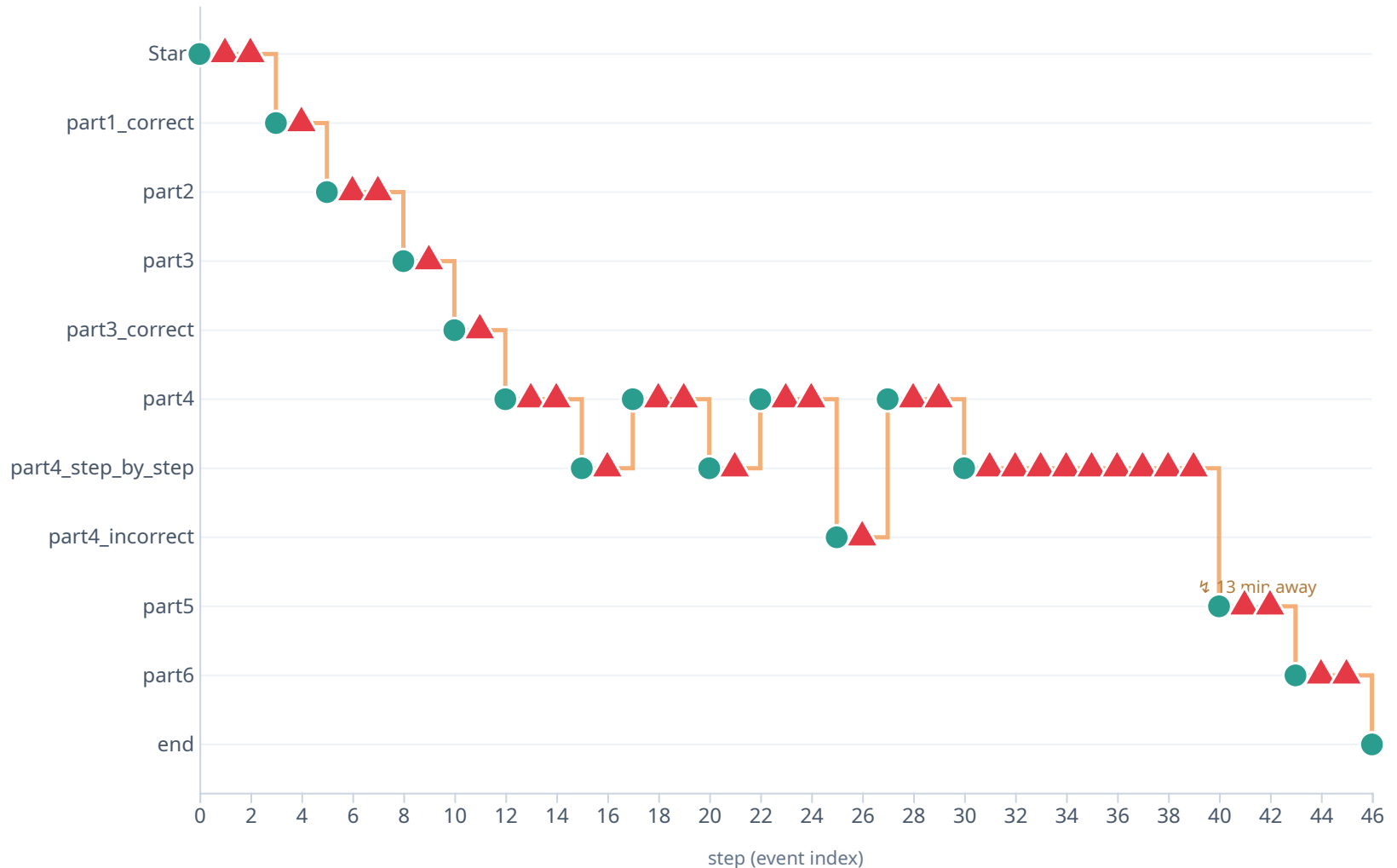
[Reveal hint](#)

[Show me step-by-step](#)

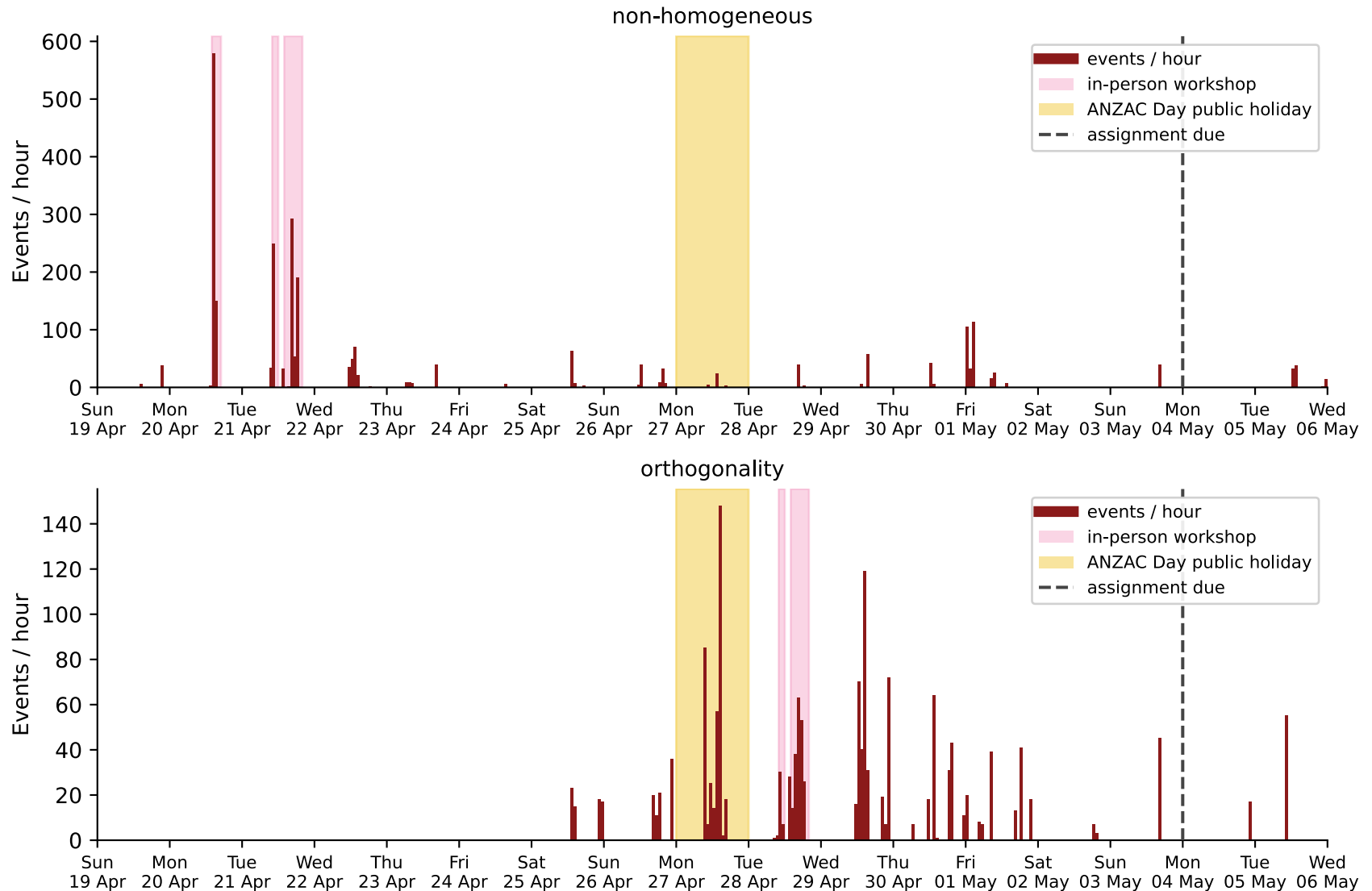
[I'm ready to answer](#)

1. What did we do?

One student's journey

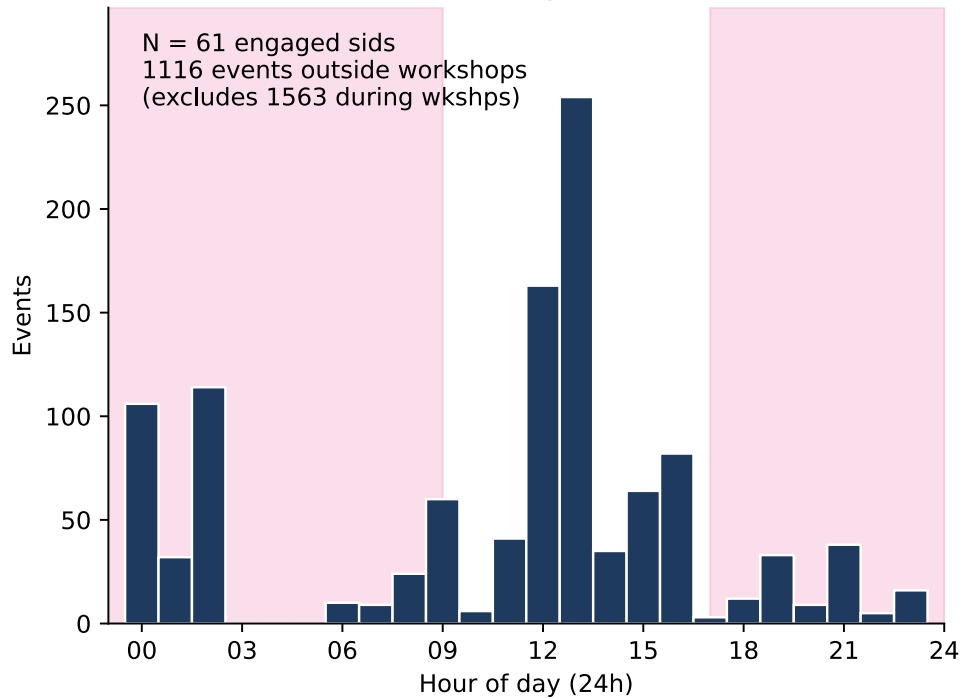


2. Who did we reach?

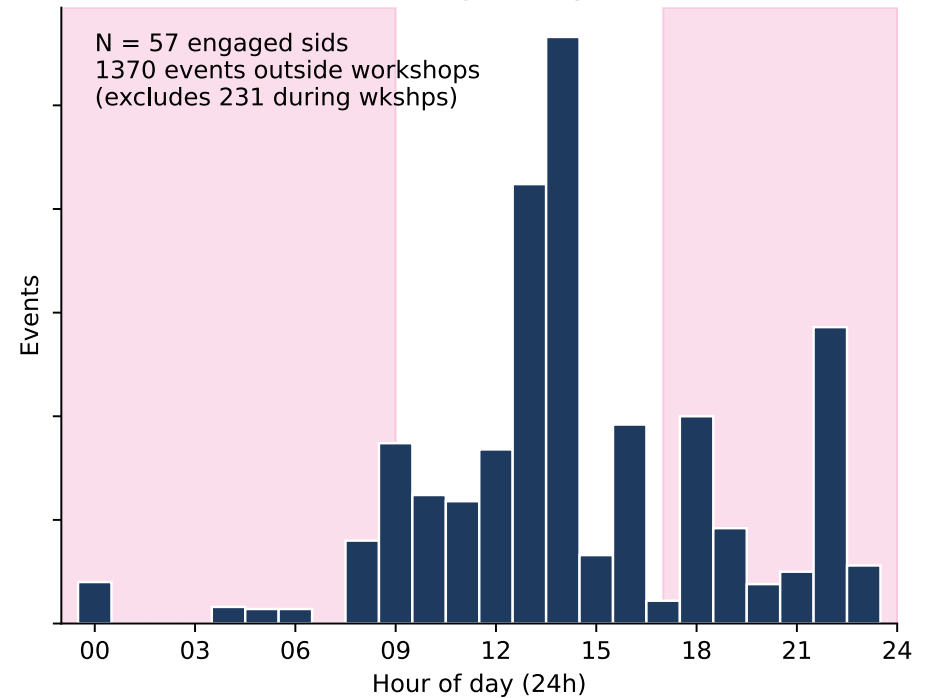


2. Who did we reach?

non-homogeneous



orthogonality



3. Did it work?

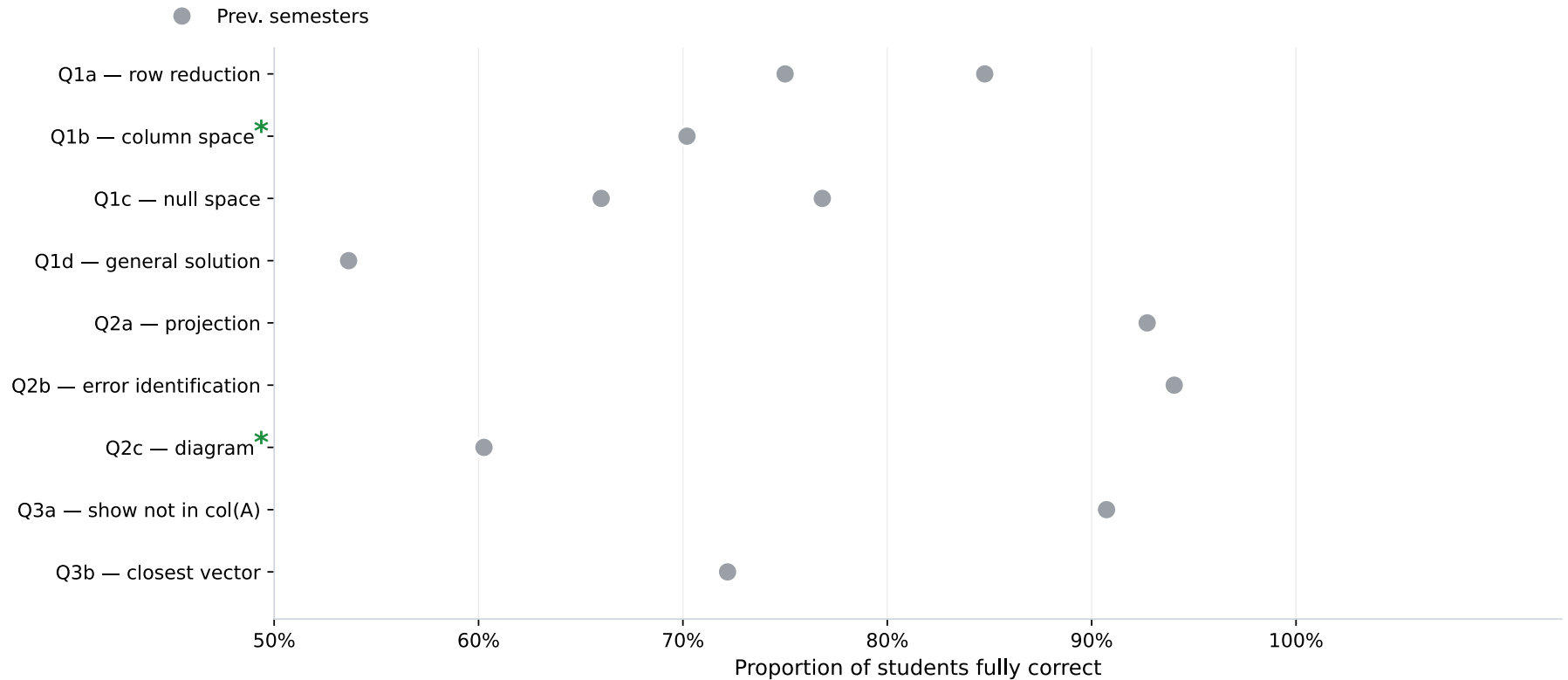
“I was just emailing you to let you know that I have found the workshop hints to be super informative, intuitive and helpful. It would be awesome if the teaching staff could produce them on a weekly basis.”

— student email to Unit Coordinator

“It read my mind”

“Better than Khan Academy”

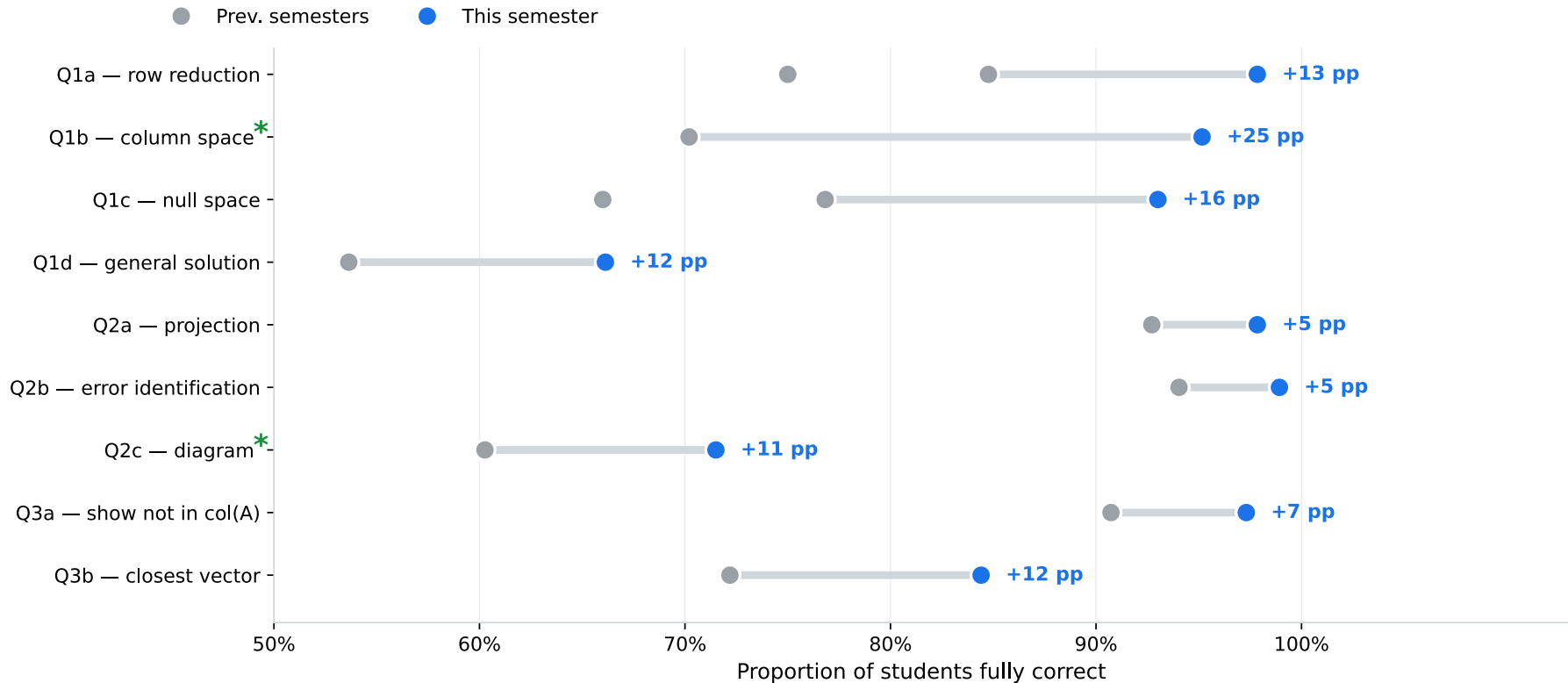
3. Did it work?



3. Did it work?

Correctness improved on every question

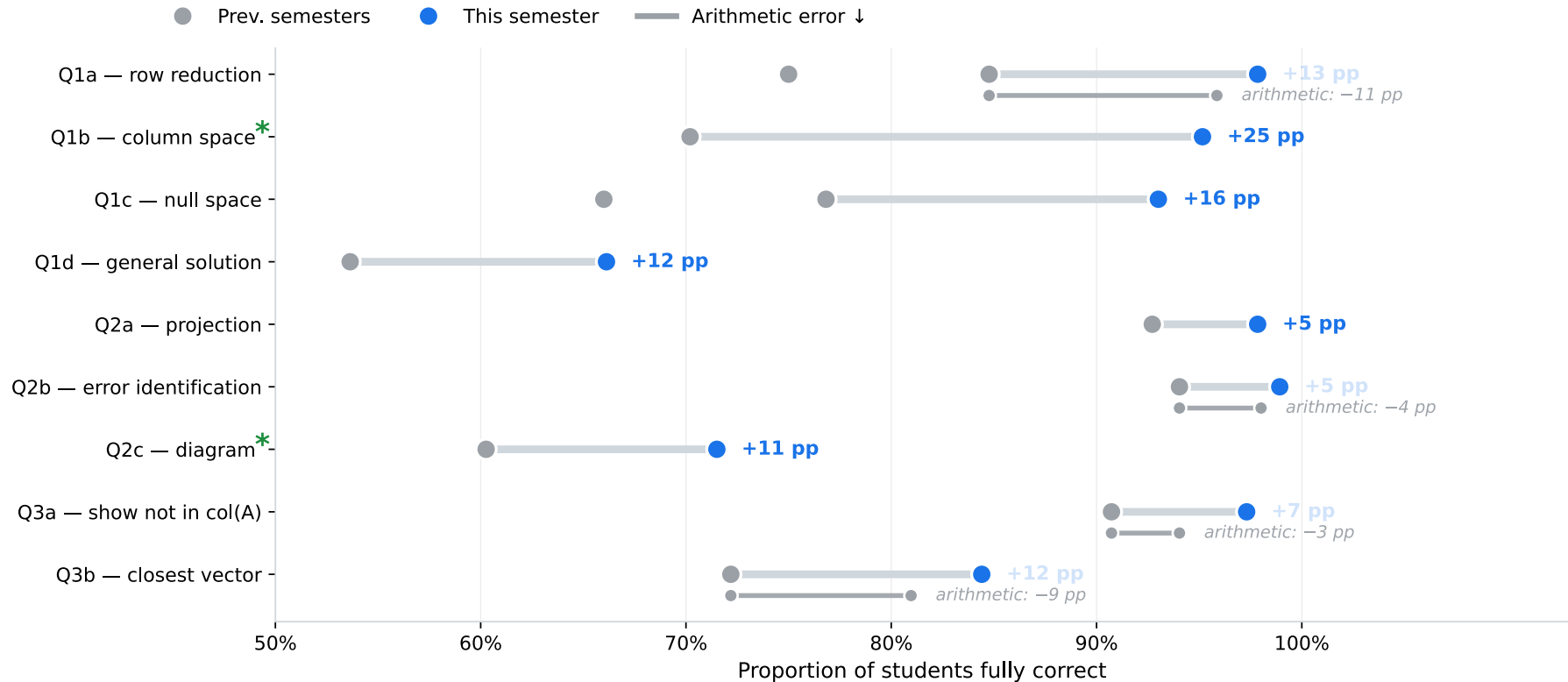
Average improvement: +11.9 percentage points (95% CI: +7.0 to +16.7 pp).
All +pp gains: BH $p < 0.03$.



3. Did it work?

Correctness improved on every question

Average improvement: +11.9 percentage points (95% CI: +7.0 to +16.7 pp).
All +pp gains: BH $p < 0.03$.



3. Did it work?

Correctness improved on every question

Average improvement: +11.9 percentage points (95% CI: +7.0 to +16.7 pp).

All +pp gains: BH $p < 0.03$.

Sub-bars show dominant mechanism: * BH $p < 0.05$, ** BH $p < 0.01$.

'Missing explanation' effects also had significant 'concept right' gain (excludes arithmetic errors)

